Post-Implementation Power Optimization of a Clock-Gated ALU using FPGA Fabric

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Redundant switching activity across non-participating functional units dominates dynamic power dissipation in parallel computing blocks. This project addresses this by developing a 32-bit low-power hierarchical ALU, utilising architectural clock gating to keep unused inputs from switching. Unlike a standard ALU, where all sub-modules may perform computations simultaneously, the proposed architecture uses direct combinational opcode decoding with hardware clock buffers to deactivate the clock trees of idle blocks. This eliminates redundant register updates and transistor switching. Validation was performed using Vivado's post-implementation signoff methodology, with a high-stress SAIF toggle profile annotated on the physical netlist at an operating frequency of 50 MHz. The proposed architecture achieved a 17.65% reduction in dynamic power dissipation compared to a baseline ALU (from 17 mW to 14 mW), with core logic dissipation reduced from 3 mW to 1 mW.

1 Theoretical Foundations of Clock-Gated Architectures

1.1 Dynamic Power Expression

The fundamental physical behavior of a CMOS circuit can be evaluated through the classic dynamic power dissipation equation:

$$P_{\text{dynamic}} = \alpha \cdot C_{\text{eff}} \cdot V_{dd}^2 \cdot f \tag{1}$$

Where:

- α represents the switching activity factor ($0 \leq \alpha \leq 1$), defining the average number of power-consuming logic transitions per clock cycle.
- C_{eff} represents the total parasitic load capacitance of the physical routing traces and transistor gates.
- V_{dd} represents the operating supply voltage.
- f represents the clock frequency.

Scaling down the global clock frequency (f) would directly penalise the overall system throughput, so it is not a practical way to reduce dynamic power consumption. Further, once a design has been realised, the capacitance is fixed, and hardware constraints

limit the reduction in supply voltage (V_{DD}). Although the global clock frequency cannot be changed, modern low-power architectures can instead manipulate frequency locally. This project aims to verify this method of dynamic power reduction by driving the local clock frequency of unselected sub-modules to zero ($f_{\text{local}} = 0$ Hz). This way, both the effective local frequency and the switching activity factor (α) are reduced to zero, eliminating dynamic power dissipation in unselected sub-modules without affecting the system throughput.

1.2 Standard ALU Architectures

In a conventional hierarchical 32-bit ALU, the global clock tree distributes clock pulses to all functional processing blocks simultaneously via shared distribution networks. When each active clock edge arrives, every sub-module performs calculations simultaneously, regardless of the selected instruction.

For instance, if a simple bitwise AND operation is to be evaluated, the dedicated clock branch feeding the 32-bit Multiplier still toggles, continuously charging and discharging internal parasitic capacitances along its clock path and internal logic. This continuous oscillation within the unselected sub-module results in significant dynamic power dissipation, even though its outputs are immediately discarded. This results in an unnecessarily high global switching activity factor (α).

1.3 Architectural Clock Gating

The core microarchitectural solution implemented in this project is Clock Gating. The objective is to decouple idle execution blocks from the active clock distribution network. By decoding the incoming opcode combinationally within the arrival cycle, the system identifies the target functional unit. It isolates the clock feeds of the remaining sub-modules using dedicated clock-gating buffers (`BUFGCE`).

In a conventional architecture, the clock network continuously distributes an active frequency f to every internal register, regardless of whether its operation is selected. Under the proposed clock-gating scheme, the clock buffer for any unselected module i is disabled, thereby setting its local clock frequency to zero ($f_i = 0$ Hz).

This prevents the sub-module's flip-flops from receiving a clock edge, causing them to hold their previous states. Because the register outputs never change, the internal logic remains completely static, eliminating all dynamic power dissipation in the unselected unit.

2 Microarchitectural Hardware Design (RTL)

2.1 Structural Design Hierarchy

To evaluate the power characteristics of architectural clock gating, a highly modular, decoupled microarchitecture was designed and implemented. The design is partitioned across five structural files to isolate functional execution domains into independently gated clock zones.

- `alu_pkg.vhd`: A centralised design library containing global constants, bus width definitions (`DATA_WIDTH = 32`, `OP_WIDTH = 4`), and the enumerated opcode matrix for all 16 arithmetic, logic, and shifting operations.
- `arith_unit.vhd`: This block encapsulates synchronous 32-bit addition, subtraction, and a physical 32-bit unsigned multiplier.
- `logic_unit.vhd`: A bitwise execution block performing synchronous Boolean functions (`AND`, `OR`, `XOR`, `NOT`) across the 32-bit operand inputs.
- `shift_unit.vhd`: A dedicated barrel shifter capable of multi-bit arithmetic and logical left/right shifting operations.
- `top_alu.vhd`: The structural top-level wrapper that instantiates the sub-modules and establishes the internal interconnect networks. In the low-power variant of the design, this entity embeds the combinational opcode-decoding logic alongside the dedicated hardware clock-buffer primitives (`BUFGCE`) that control the regional clock feeds to each sub-module.

2.2 Architectural Clock Control

In the baseline ALU, the global clock net is routed directly to the clock pins of all execution blocks. In contrast, the low-power ALU architecture intercepts the clock distribution network by routing clock feeds through dedicated hardware clock-gating primitives within the `top_alu.vhd` wrapper.

The gating control logic operates via a direct combinational opcode-decoding scheme. When data arrives at the primary inputs, the 4-bit opcode bus is decoded combinatorially to generate independent enable signals (CE).

For example, if the opcode belongs to the Logic Class:

- $CE(\text{Logic_Unit}) = 1$, meaning local clock = global clock
- $CE(\text{Arith_Unit}) = 0$, meaning local clock = 0 Hz
- $CE(\text{Shift_Unit}) = 0$, meaning local clock = 0 Hz

By directly controlling clock distribution at the regional buffer primitives (`BUFGCE`), the architecture isolates the clock-tree branches feeding idle domains. Because unselected sub-modules receive no active clock transitions, their internal input registers hold their previous state without toggling. Therefore, internal gates remain perfectly static during the operation, eliminating dynamic power dissipation.

3 Simulation-Driven Verification Methodology

3.1 The High Stress Test Workload

To evaluate the power characteristics of the clock-gated ALU under worst-case conditions, the testbench (`tb_top_alu.vhd`) applies a dedicated High Stress Test Workload rather than an incrementing counter.

The stimulus architecture operates via a dual-loop pattern:

- **Opcode Sweep:** An inner loop iterates sequentially through all 16 operations (arithmetic, logic, and shift), activating each functional execution block one after another.
- **Workload Iteration:** An outer loop repeats this 16-operation sequence 10 times consecutively, concluding the active workload at 6,460 ns across a total simulation timeline of 6,500 ns.

To maximize switching activity at the primary inputs, the operand data buses undergo complete bitwise inversions on every operation pass. Each test vector is held for two clock cycles (40 ns) to accommodate pipeline latency, alternating back-to-back between `x"55555555"` and `x"AAAAAAAA"`:

$$\text{Operation } k : \begin{cases} src_A = \texttt{x"55555555"} \\ src_B = \texttt{x"AAAAAAAA"} \end{cases} \implies \text{Operation } k{+}1 : \begin{cases} src_A = \text{NOT } src_A \\ src_B = \text{NOT } src_B \end{cases} \tag{2}$$

This forces all 32 bits of both input buses to flip simultaneously across consecutive operations. This high-stress stimulus provides an ideal benchmark for evaluating clock gating: even while the external data buses toggle aggressively, the clock-gated idle sub-modules receive no active clock transitions, keeping their internal registers and logic completely static.

3.2 Post-Implementation SAIF Generation

To transition the power analysis from an approximation into a robust verification, a deterministic tool flow was developed. Rather than relying on Vivado's default Vectorless Estimation, this project records the explicit physical switching behavior of the layout via a *Switching Activity Interchange Format (SAIF)* pipeline.

The execution pipeline transitions systematically through three stages:

1. **Physical Placement and Routing:** The synthesized VHDL netlist is fully implemented on the target Artix-7 silicon die, generating a timing-accurate physical gate representation where all internal signal traces are bounded to real hardware coordinates.
2. **Post-Implementation Functional Simulation:** The physical netlist is loaded into the Vivado simulator alongside the testbench. This configuration mimics the exact cycle-by-cycle electrical behavior of a manufactured chip.
3. **SAIF Extraction:** Before running the simulation, the Tcl console initializes a logging instance targeting the hierarchical Unit Under Test (`UUT`). The specific operational sequence is executed as follows:

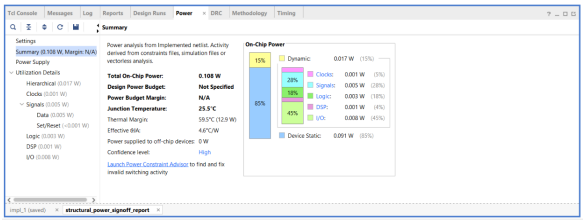
```
restart
set proj_dir [get_property DIRECTORY [current_project]]
open_saif "$proj_dir/switching_activity.saif"
log_saif [get_objects -r /tb_top_alu/UUT/*]
run 6460 ns
close_saif
```

The recursive tracking flag (`-r`) commands the simulation engine to monitor not just the top-level boundary ports, but all internal interconnections, flip-flops, and dedicated primitives within the instantiated design hierarchy. As the 6,460 ns stress test executes, every physical toggle event is recorded and compiled into the `switching_activity.saif` file.

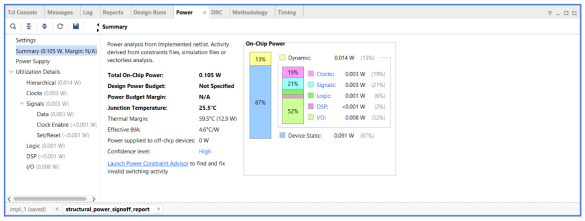
4 Experimental Results

4.1 Dynamic Consumption Analysis

Post-implementation power characterization was conducted by comparing the baseline standard ALU against the clock-gated ALU. Both designs were analyzed under the identical 6,460 ns high stress test workload at an operating frequency of 50 MHz, utilizing 100% complete post-simulation SAIF back-annotation. Table 1 establishes the definitive data comparison matrix between the two design variants.



(a) Standard ALU Power Report.



(b) Clock-Gated ALU Power Report.

Figure 1: Post-Implementation Power Summaries showcasing the distribution of power dissipation inside the Vivado Design Suite.

Table 1: Post-Implementation Power Comparison Matrix at 50 MHz

Power Component	**Standard ALU**	**Clock-Gated ALU**	**Delta**
Total On-Chip Power	108 mW	105 mW	−2.78%
Device Static Power	91 mW	91 mW	0.00%
Dynamic Power	17 mW	14 mW	**−17.65%**
Dynamic Breakdown:			
Clocks	1 mW	3 mW	+200.00%
Internal Signals	5 mW	3 mW	−40.00%
Internal Logic	3 mW	1 mW	−66.67%
I/O	8 mW	8 mW	0.00%

Static power is fixed at 91 mW because it is strictly determined by the physical transistor leakage of the underlying 28 nm silicon substrate. RTL-level modifications cannot adjust static leakage; it remains active as long as the chip is powered. The true metric of design optimization must be tracked inside the **Total Dynamic Power** layer. Under the high stress workload, the baseline design dissipates 17 mW of dynamic power. Implementing regional clock gating using dedicated `BUFGCE` primitives drives this metric down to 14 mW, achieving a noteworthy **17.65% reduction in dynamic power**.

This saving is explicitly validated by the internal breakdowns: *Internal Logic* dropped by 66.67% (from 3 mW to 1 mW) and *Internal Signals* fell by 40.00% (from 5 mW to 3 mW). While the dedicated clock distribution tree power increased from 1 mW to 3 mW due to regional `BUFGCE` buffer overhead, the substantial savings across logic and interconnect nets outweigh this penalty, yielding a net reduction in overall dynamic dissipation under sustained continuous switching.

5 Conclusion

5.1 Summary of Achievements

This project successfully validates architectural clock gating using dedicated `BUFGCE` primitives as an effective power-mitigation strategy for a modular 32-bit ALU. Evaluated under a sustained 6,460 ns high-stress workload at 50 MHz, the gated architecture achieved a **17.65%** net reduction in dynamic power compared to the baseline design, primarily driven by a **66.67%** drop in internal logic dissipation and a **40.00%** decrease in signal routing power. By selectively disabling clock distribution to dormant sub-modules, unnecessary switching activity is suppressed at the hardware level without impacting operational throughput. Furthermore, post-implementation timing analysis confirmed robust timing closure with a positive Worst Negative Slack (WNS) of **+6.537 ns** (as illustrated in Figure 2), demonstrating that significant dynamic power savings can be realized on 28 nm FPGA fabric without incurring timing penalties or requiring complex pipelining overhead.

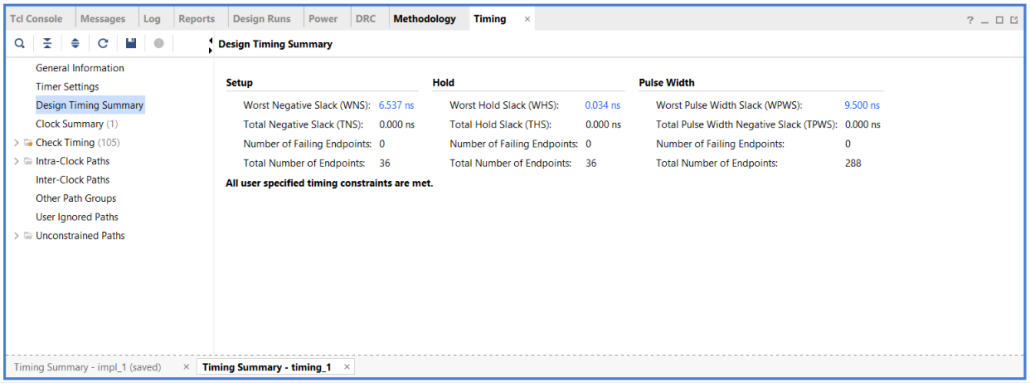


Figure 2: Post-implementation timing summary of the clock-gated ALU at 50 MHz, confirming timing closure with a Worst Negative Slack (WNS) of +6.537 ns.